Introduction to University Mathematics

General Information.
Discipline: Mathematics *Course code*: 201-SNU-AB
Ponderation: 2-2-2 *Credits*: 2 *Prerequisite*: 201-SN2-RE
Objective: 0GNF: Consolidate one's scientific culture in a scientific field.

- Develop a clear understanding of the mathematical proof and prepare for university level mathematics.

> Your teacher will give you their schedule and availability.
> *Students are strongly advised to seek help from their instructor as soon as they encounter difficulties in the course.*

Introduction. This science option course will provide an introduction to the fundamental concepts in university-level mathematics. The primary purpose is to learn how to think logically and mathematically and to understand the importance of the mathematical proof. To achieve this goal, this class will touch on basic logic and set theory, which are foundational to all areas of mathematics. Students will learn to work with logical statements, understand set operations, and use these tools to construct clear and rigorous proofs. The course will then also cover divisibility and congruence, which play a crucial role in number theory and algebra.

Two of the following three key topics will also be covered, at the discretion of the teacher: basics of group theory, real analysis, and complex numbers. Where possible, this class will also examine the applications of these topics to other sciences.

At the end of this class, the student will have a clear understanding of the mathematical proof and a better idea of what to expect from university math courses.

Teaching Methods. This course will be 60 hours, meeting three times a week for a total of 4 hours a week. Most teachers of this course rely mainly on the lecture method, although most also employ at least one of the following techniques as well: question-and-answer sessions, labs, problem solving periods and class discussions. Generally, each class session starts with a question period of previous topics, then new material is introduced, followed by worked examples. No marks are deducted for absenteeism; however, failure to keep pace with the lectures generally results in a cumulative inability to cope with the material and a possible failure in the course. A student will generally succeed or fail depending on how many problems have been attempted and solved successfully. It is entirely the student's responsibility to complete suggested homework assignments as soon as possible following the lecture, as the material will be fresher in their mind. This also allows the student the maximum benefit from any discussion of the homework (which usually occurs in the following class). Each teacher will provide supplementary notes and problems as he/she sees fit.

Reference Text. While there is no set textbook for this class, your teacher may refer you to certain textbooks that may help with your understanding of the material.

Evaluation Plan.

- Final Exam: 30%
- Other tests 20%
- Other assessments: 50%

The final exam in combination with the other tests constitute the final assessment. The Final Exam will be 90 minutes, written during the final exam period. Other assessments are at the teacher's discretion, but will be specified in an appendix to this outline produced by the teacher. These may include quizzes, written assignments, group work, etc. Your teacher will provide you with an addendum to this outline that describes these other assessments.

Course Costs.
There are no extra costs.

Other Resources.
Math Website.
http://departments.johnabbott.qc.ca/departments/mathematics

Math Study Area. Located in H-200A and H-200B; the common area is usually open from 8:30 to 17:30 on weekdays as a quiet study space. Computers and printers are available for math-related assignments. It is also possible to borrow course materials when the attendant is present.

Math Help Centre. Located near H-211; teachers are on duty from 8:30 until 15:30 to give math help on a drop-in basis.

Academic Success Centre. The Academic Success Centre, located in H-139, offers study skills workshops and individual tutoring.

College Policies.
Policy No. 7 - IPESA, Institutional Policy on the Evaluation of Student Achievement: https://www.johnabbott.qc.ca/wp-content/uploads/2021/05/Policy-No.-7-IPESA-FINAL.pdf.

Changes to Evaluation Plan in Course Outline (Article 5.3). Changes require documented unanimous consent from regularly attending students and approval by the department and the program dean.

Religious Holidays (Article 3.2.13 and 4.1.6). Students who wish to miss classes in order to observe religious holidays must inform their teacher of their intent in writing within the first two weeks of the semester.

Student Rights and Responsibilities: (Article 3.2.18). It is the responsibility of students to keep all assessed material returned to them and/or all digital work submitted to the teacher in the event of a grade review. (The deadline for a Grade Review is 4 weeks after the start of the next regular semester.)

Student Rights and Responsibilities: (Article 3.3.6). Students have the right to receive graded evaluations, for regular day division courses, within two weeks after the due date or exam/test date, except in extenuating circumstances. A maximum of three (3) weeks may apply in certain circumstances (ex. major essays) if approved by the department and stated on the course outline. For evaluations at the end of the semester/course, the results must be given to the student by the grade submission deadline (see current Academic Calendar). For intensive courses (i.e.: intersession, abridged courses) and AEC courses, timely feedback must be adjusted accordingly.

Academic Procedure: Academic Integrity, Cheating and Plagiarism (Article 9.1 and 9.2). Cheating and plagiarism are unacceptable at John Abbott College. They represent infractions against academic integrity. Students are expected to conduct themselves accordingly and must be responsible for all of their actions. For more information, see the Academic Procedure on Academic Integrity.

College definition of Cheating: Cheating means any dishonest or deceptive practice relative to examinations, tests, quizzes, lab assignments, research papers or other forms of evaluation tasks. Cheating includes, but is not restricted to, making use of or being in possession of unauthorized material or devices and/or obtaining or providing unauthorized assistance in writing examinations, papers or any other evaluation task and submitting the same work in more than one course without the teacher's permission. It is incumbent upon the department through the teacher to ensure students are forewarned about unauthorized material, devices or practices that are not permitted.

College definition of Plagiarism: Plagiarism is a form of cheating. It includes copying or paraphrasing (expressing the ideas of someone else in one's own words), of another person's work or the use of another person's work or ideas without acknowledgement of its source. Plagiarism can be from any source including books, magazines, electronic or photographic media or another student's paper or work.

Policy No. 13 - Policy on Student Conduct and Discipline Procedures: https://www.johnabbott.qc.ca/wp-content/uploads/2021/09/POLICY-13_-_policy_on_student_conduct-Final-BOG-2021-09-21.pdf.

Course Content. The teacher will cover the first three topics and then choose two of the topics from Option 1, Option 2, and Option 3. The exercises listed below form a good basis for homework and should help you practice and learn the material taught in this course. Your teacher may supplement this list during the semester. Regular work done as the course progresses should make it easier for you to master the course.

Logic and Proofs: Worksheet 1

Set Theory: Worksheet 2

Divisibility and Congruences: Worksheet 3

Real Analysis: (Option 1) Worksheet 4

Group Theory: (Option 2) Worksheet 5

Complex Numbers: (Option 3) Worksheet 6

OBJECTIVES	STANDARDS
Statement of the competency Develop a clear understanding of the mathematical proof and prepare for university level mathematics.	
Elements of the Competency	**General Performance Criteria**
1. Logic and Proofs	1.1 propositions and mathematical language 1.2 proof vs. evidence
2. Set Theory	2.1 basic notions 2.2 induction
3. Divisibility and Congruences	3.1 divisibility 3.2 congruences
4. Real Analysis	4.1 limits 4.2 continuity and differentiation
5. Group Theory	5.1 group axioms and abelian groups 5.2 group order and symmetry 5.3 subgroups 5.4 applications (optional)
6. Complex Numbers	6.1 basic notions 6.2 polar form 6.3 applications (optional)

Specific Performance Criteria	Intermediate Learning Objectives
7. *Understanding Logic and Proofs*	
7.1 Propositions and Mathematical Language	7.1.1. determine whether an assertion is true or false 7.1.2. understand conditionals 7.1.3. write common concepts in math (for example, even numbers can be written as $2n$ and odds as $2n+1$ 7.1.4. use tautologies and conditionals 7.1.5. understand the difference between if and iff
7.2 Proof vs. Evidence	7.2.1. develop a solid and systematic approach to the logic underlying proofs 7.2.2. understand the importance of a counterexample 7.2.3. prove logic based theorems and lemmas by direct proof 7.2.4. prove logic based theorems and lemmas by contradiction
8. *Set Theory*	
8.1 Basic Notions	8.1.1. understand the concept of a set 8.1.2. understand basic notations and definitions 8.1.3. use set operations to obtain a combination of elements as per the operation performed on them 8.1.4. prove that two sets are equal or that a set is empty 8.1.5. understand definition and ordering of natural numbers
8.2 Mathematical Induction	8.2.1. understand the theory behind the principle of mathematical induction 8.2.2. prove certain theorems by induction
9. *Divisibility and Congruence*	
9.1 Divisibility	9.1.1. use the division algorithm 9.1.2. identify prime and relatively prime numbers 9.1.3. express an object as a product of its prime factors 9.1.4. understand Euclid's Theorem 9.1.5. apply the Euclidean Algorithm 9.1.6. find the greatest common divisors and the lowest common multiples of a math object
9.2 Congruences	9.2.1. understand the meaning of $\mathbb{Z}_n$, the integers modulo n 9.2.2. find inverses in $\mathbb{Z}_n$ 9.2.3. solve systems of equations in $\mathbb{Z}_n$ 9.2.4. apply Fermat's Theorem
10. *Real Analysis*	
10.1 Limits	10.1.1. directly use the ϵ definition of the limit of a sequence 10.1.2. directly use the ϵ-δ definition of the limit of a function
10.2 Continuity and Differentiation	10.2.1. prove the continuity or discontinuity of a function in terms of ϵ-δ 10.2.2. determine differentiability of a function in terms of ϵ-δ
11. *Group Theory*	
11.1 Group Axioms and Abelian Groups	11.1.1. understand whether or not something is a group 11.1.2. prove whether or not something is a group using the axioms 11.1.3. apply definition of a homomorphism 11.1.4. prove whether or not a set is an abelian group
11.2 Group Order and Symmetry	11.2.1. understand the concept of symmetry 11.2.2. define and find the order of a group 11.2.3. find the generating set of a group
11.3 Subgroups	11.3.1. define a subgroup 11.3.2. apply Lagrange's Theorem
11.4 Applications (optional)	11.4.1. understand the connection between cyclic groups and music theory 11.4.2. solve cryptography problems by using permutation groups and substitution permutation networks 11.4.3. understand the symmetry of molecules and chemistry
12. *Complex Numbers*	
12.1 Basic Notions	12.1.1. apply addition, subtraction, multiplication, conjugation, division of complex nuumbers 12.1.2. identify the real and imaginary parts of a complex number 12.1.3. find the modulus of a complex number 12.1.4. interpret complex numbers geometrically
12.2 Polar Form	12.2.1. find the argument of a complex number 12.2.2. apply Euler's Formula 12.2.3. find roots of unity
12.3 Applications (optional)	12.3.1. understand the connection between complex numbers and signal processing: Fourier Transform 12.3.2. electrical engineering: perform circuit analysis

Specific Performance Criteria	Intermediate Learning Objectives